**CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT**

Course Code and Title**: S-CSPC212 – DISCRETE STRUCTURES**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

**DIRECTIONS:** Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, as you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

| Format, parts, and description | Compliance | |
|---|---|---|
| | Complied With | Not Complied With |
| **Format** | | |
| 1. The 2023-2024 syllabus format is used. | / | |
| 2. The font used in the syllabus preparation is Times New Roman (11). | / | |
| 3. DLSUD (being first) and college logos are beside each other at the top (center). | / | |
| 4. Preliminaries include all the contact details of the professor, such as the day and time. | / | |
| 5. Each WEEK has a corresponding date. | / | |
| **Content** | | |
| 6. The Learning Outcomes have more HOTS than LOTS. | / | |
| 7. CLOs and TLOs are identified for each module/lesson/topic. | / | |
| 8. Gospel reading and reflection precede the presentation of each topic under the teaching-learning activities. | / | |
| 9. The journal articles/references are within the last five years. | / | |
| 10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies. | / | |
| 11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students. | / | |
| **Assessments** | | |
| 12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus. | / | |
| 13. Formative Assessments are not reflected anymore in the syllabus. | / | |
| 14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities. | / | |
| 15. The syllabus presents one-module(lesson)-one-assessment correspondence. | / | |
| 16. Only one Enabling Assessment is reflected in a given week | / | |
| 17. The Enabling Assessments are aligned with the identified CLOs and TLOs. | / | |
| 18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term. | / | |
| 19. Thirty percent (30%) is allotted for Summative Assessments per term. | / | |
| 20. Twenty percent (20%) is allotted for Class Participation/Attendance per term. | / | |

| | | |
|---|---|---|
| 21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade. | / | |
| 22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities. | / | |
| 23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities. | / | |
| 24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD | / | |
| 25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD. | / | |
| **References and Curriculum Map** | | |
| 26. The References must be within the last 5 years except for canons and seminal studies. | / | |
| 27. It must be a combination of books, eBooks, journal articles, videos, etc. | / | |
| 28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS. | / | |
| 29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs) | / | |

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

**Self-reviewed:**

Ms. Marnelli B. Erquita
Faculty

**Checked, verified, and endorsed:**

***Rolando B. Barrameda***
Dept. Curriculum Committee Reviewer

**Checked, verified, and approved:**

***Josephine T. Eduardo***
Department Chair

***Dr. Rubie M. Causaren***
College Dean

**De La Salle University - Dasmariñas**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER SCIENCE DEPARTMENT**

## COURSE SYLLABUS

| | | |
|---|---|---|
| **COURSE CODE** | **:** | S-CSPC212 |
| **COURSE TITLE** | **:** | DISCRETE STRUCTURES |
| **COURSE TYPE** | **:** | Lecture |
| **COURSE CREDIT** | **:** | 3 Units |
| **LEARNING MODALITY** | **:** | Traditional Blended Learning Model |
| **PRE-REQUISITES** | **:** | None |
| **CO-REQUISITES** | **:** | None |
| **PROFESSOR** | **:** | MS. MARNELLI B. ERQUITA |
| **CONTACT DETAILS** | **:** | mberquita@dlsud.edu.ph/CH(M/T - 09:00-10:00, W - 10:00-12:00, F – 08:00-09:00) |

**COURSE DESCRIPTION:**

This course covers fundamentals of logic and sets, fundamental principles of counting, algorithms, and some concepts in graph theory.

**COURSE LEARNING OUTCOMES:**

By the end of this course, students can:

CLO1. evaluate different approaches in addressing different real-world problems using the mathematical concepts.
CLO2. assess the significance and impact of the concepts learned and how these concepts can solve mathematical problems that we encounter in our everyday lives.
CLO3. apply mathematically correct terminology and notation.
CLO4. evaluate the validity of an argument.
CLO5. understand the basics of algorithm analysis, including big-O notation.
CLO6. remember how to work with discrete structures such as sets, graphs, and trees, which are abstract mathematical structures used to represent discrete objects and the relationships between these objects.
CLO7. demonstrate 21st-century skills in all learning activities.

**LEARNING PLAN**

**Learning Outcomes for Midterm Period**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO1. Evaluate different approaches in addressing different real-world problems using the mathematical concepts learned. | TLO1. Understand the concept of propositions.<br>TLO2. Determine the truth value of a proposition or compound proposition.<br>TLO3. Learn to write a mathematical proof. |
| CLO2. Assess the significance and impact of the concepts learned and how these concepts can solve mathematical problems that we encounter in our everyday lives. | TLO4. Learn to read, comprehend, and construct mathematical arguments.<br>TLO5. Exhibit knowledge of how to work with discrete structures. |

| | |
|---|---|
| CLO4 Evaluate the validity of an argument. | TLO9. Apply standard logical identities/equivalences and truth tables to demonstrate logical equivalency, tautology, contradiction, and contingency of logical statements.<br>TLO10. Construct valid arguments.<br><br>*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-solving* |

| **Module No and Learning Outcomes** | **Week** | **Teaching-Learning Activities / Assessment Strategy** | **Technology Enabler** | **Asynchronous** | **Face-to-Face** | **Allotted Hours** |
|---|---|---|---|---|---|---|
| Module 0 | 1 (Aug 14-19) | • Gospel Reading and Reflection<br>• Presentation of Module 0: Course Overview/Introduction<br>• Getting-to-know: Teacher and Students<br>• Presentation of Module 1- Propositional Logic<br>• CP: Assignment No. 1 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| Module 1<br><br>CLO1<br>TLO1<br>TLO2<br>TLO3 | 2 (Aug. 21-26) | • Gospel Reading and Reflection<br>• Module 1: Propositional Logic<br>• Self-paced reading (Advanced Reading)<br>• EA: Seatwork No. 1 | Schoolbook | / | | 3.0 |
| | 3 (Aug 28 – Sept 2) | • Gospel Reading and Reflection<br>• Presentation of Module 1: Propositional Logic (Discussion)<br>• EA: Short Quiz No. 1 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| Module 2<br>Module 3<br>CLO2<br>TLO4<br>TLO5 | 4 (Sept 4-9) | • Gospel Reading and Reflection<br>• Module 2: Propositional Equivalences<br>• Module 3: Predicates and Quantifiers<br>• Self-paced reading (Advanced Reading)<br>• CP: Assignment No. 2 | Schoolbook | / | | 3.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 5 (Sept 11-16) | • Gospel Reading and Reflection<br>• Presentation of Module 2 – Propositional Equivalences and Module 3 - Predicates and Quantifiers (Discussion)<br>• EA: Short Quiz No. 2 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 6 (Sept 18-23) | • Gospel Reading and Reflection<br>• Module 4: Rules of Inference<br>• Self-paced Reading | Schoolbook | / | | 3.0 |
| Module 4 CLO4 TLO9 TLO10 | 7 (Sept 25 – 30) | • Gospel Reading and Reflection<br>• Midterm Long Quiz<br>• Coverage Modules 1,2, and 3<br>• Presentation of Module 4 – Rules of Inference | Pen and Paper Test<br><br>MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 8 (Oct 2 - 7) | • Gospel Reading and Reflection<br>• Continuation of Module 4 – Rules of Inference<br>• EA: Seatwork No. 2 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 9 (Oct 9-14) | **MIDTERM EXAMINATION** | | | / | 3.0 |
| | | | ***TOTAL*** | ***9.0*** | ***18.0*** | ***27.0*** |

**Learning Outcomes for Final Period**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO3. Apply mathematically correct terminology and notation. | TLO6. Exhibit knowledge of how to work with discrete structures.<br>TLO7. Develop representation of discrete objects and relationships between these objects.<br>TLO8. Develop your own mathematical models. |
| CLO5. Understand the basics of algorithm analysis, including big-O notation. | TLO11. Know the growth of a function.<br>TLO12. Know the complexity of an algorithm. |
| CLO6. Remember how to work with discrete structures such as sets, graphs, and trees, which are abstract mathematical structures used to represent discrete objects and the relationships between these objects. | TLO13. Learn the importance of sets and how it is used to build discrete structures such as sets, graphs, and trees. |
| CLO7. Demonstrate 21st-century skills in all learning activities. | TLO14. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of the topic, but integrated with assessments) |

| Module No and Learning Outcomes | Week | Teaching-Learning Activities / Assessment Strategy<br>Module Activities | Technology Enabler | Asynchronous | Face-toFace | Allotted Hours |
|---|---|---|---|---|---|---|
| | 10 (Oct 16-21) | **STUDENT'S WEEK** | | / | | 3.0 |
| | 11 (Oct 23-28) | **INSTITUTIONAL SELF-CARE WEEK** | | / | | 3.0 |
| Module 5<br>CLO3<br>CLO6<br>TLO6<br>TLO7<br>TLO8<br>TLO13 | 12 (Oct 30-Nov 4) | • Gospel Reading and Reflection<br>• Module 5: Set and Set Operations<br>• Self-paced Reading<br>• CP: Assignment No. 3 | Schoolbook | / | | 3.0 |
| | 13 (Nov 6-11) | • Gospel Reading and reflection<br>• Presentation of Module 5 – Set and Set Operations<br>• EA: Seatwork No. 3 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| Module 6<br>CLO5<br>CLO7<br>TLO11<br>TLO12<br>TLO14 | 14 (Nov 13-18) | • Gospel Reading and Reflection<br>• Presentation of Module 6 – Algorithms<br>• EA: Short Quiz No. 3 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 15 (Nov 20-25) | • Gospel Reading and Reflection<br>• Continuation of Module 6<br>• EA: Final Long Quiz<br>• Coverage Modules 5 and 6 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 16 (Nov 27-Dec 2) | • Gospel Reading and Reflection<br>• Module 7 – Graphs and Trees<br>• Self-paced Reading (Advanced Reading) | Schoolbook | / | | 3.0 |
| Module 7<br>CLO6<br>CLO7<br>TLO13<br>TL14 | 17 (Dec 4 - 9) | • Gospel Reading and Reflection<br>• Presentation of Module 7: Graphs and Trees<br>• EA: Seatwork No. 4 | MS PowerPoint, Multimedia Devices (Laptop, Projector) | | / | 3.0 |
| | 18 (Dec 11-16) | **FINAL EXAMINATION** | | | / | 3.0 |
| | | | ***TOTAL*** | ***12.0*** | ***15.0*** | ***27.0*** |

**GRADING SYSTEM**

**Midterm**

| | |
|---|---|
| Enabling Assessments | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessment | - 30% |
| | 100% |

**Finals**

| | |
|---|---|
| Enabling Assessments | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessments | - 30% |
| | 100% |

**Overall Final Grade =** $\frac{\text{Midterm + Final}}{2}$

**COURSE POLICIES AND REQUIREMENTS:**

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.

d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

**REFERENCES**

**Online References**

| **Call number or e-provider** | **Reference Material** |
|---|---|
| Online | Rosen, Kenneth H. Discrete Mathematics and Its Application. 2019. 8th Ed. McGraw-Hill Higher Education. |
| YouTube Channel | Math - How to Construct a Truth Table from Compound Statements - YouTube<br>https://www.youtube.com/watch?v=esywilk1eSQ |

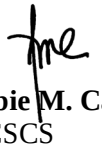
**On-Site References**

| **Provider** | **Reference Material** |
|---|---|
| QA 37.3.D63 2019 | Rosen K. 2019. Discrete Mathematics and Its Applications. Mc-Graw-Hill Education. |
| QA 39.3. D631m 2017 | Borres, M. 2017. Discrete Mathematics: An Introduction. New York, NY: Arcler Press LLC |
| QA 9 .C763 2015 | Conradie, W. & Goranko, V. 2015. Logic and Discrete Mathematics: A Concise Introduction. Chichester, West Sussex; Hoboken, NJ: John Wiley & Sons |
| QA 39.3. Ad95 2016 | Tedford, S. et.al. c2016. Advanced Discrete Mathematics. New York, NY: Magnum Publishing LLC |
| QA 39.3. R722 2013 | Rosen K. 2013. Discrete Mathematics and Its Applications, (7th ed.) New York: McGraw Hill. |

**Prepared:** **Computer Studies Department**
1st Semester 2023-2024



**Approved:** **Josephine T. Eduardo**
Chair, CSD



**Endorsed:** **Dr. Rubie M. Causaren**
Dean, CSCS

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**COMPUTER SCIENCE DEPARTMENT**

## PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

**COURSE CODE** : S-CSPC212
**COURSE TITLE** : DISCRETE STRUCTURES 1
**COURSE TYPE** : Lecture
**COURSE CREDIT** : 3 Units
**MODALITY** : **Traditional blended learning model**
**PRE-REQUISITES** : None
**CO-REQUISITES** : None

| Course Learning Outcome | Program Learning Outcomes | | | | | | |
|---|---|---|---|---|---|---|---|
| | PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 |
| 1. Evaluate different approaches in addressing different real-world problems using mathematical concepts. | ✔ | ✔ | | | ✔ | | ✔ |
| 2. Assess the significance and impact of the concepts learned and how these concepts can solve mathematical problems that we encounter in our everyday lives. | ✔ | ✔ | ✔ | | ✔ | | |
| **3.** Apply mathematically correct terminology and notation. | ✔ | ✔ | | | ✔ | | |
| 4. Evaluate the validity of an argument. | ✔ | ✔ | | | | | |
| 5. Understand the basics of algorithm analysis, including big-O notation. of arithmetic. | ✔ | | ✔ | | ✔ | | ✔ |
| 6. Remember how to work with discrete structures such as sets, graphs, and trees, which are abstract mathematical structures used to represent discrete objects and the relationships between these objects. Solve counting problems by applying counting techniques. | ✔ | ✔ | ✔ | | | | |
| 7. Demonstrate 21st-century skills in all learning activities. | | | | | | | ✔ |

*NOTE: Provide a checkmark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)*

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## GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

| Program Learning Outcomes | Graduate Attributes/ Institutional Learning Outcomes | | | | |
|---|---|---|---|---|---|
| | GA1 | GA2 | GA3 | GA4 | GA5 |
| 1. Apply concepts of computing and mathematics in different domains of computer science. | | | | ✔ | ✔ |
| 2. Identify and analyze problems and define computing requirements appropriate to their solutions. | | | | | ✔ |
| 3. Design, develop, test, and evaluate various solutions using **algorithms** for complex computing problems that take full advantage of the usability of computer resources. | | | ✔ | ✔ | ✔ |
| 4. Document computer-based systems by writing effective reports for risk-assessment impact. | | | | ✔ | ✔ |
| 5. Work effectively as an individual or as a member of a team during software development. | | | ✔ | ✔ | ✔ |
| 6. Communicate effectively with the computing community and non-computer science professionals about computing activities. | | | | ✔ | ✔ |
| 7. Recognize the legal, social, ethical, and professional issues involved in the utilization of computer technology and address these issues. | ✔ | ✔ | ✔ | | |
| 8. Engage in continuing professional development through performing independent or collaborative research as computing professionals. | | | ✔ | ✔ | ✔ |

*NOTE: Provide a checkmark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expects from its graduates.*

**GRADUATES ATTRIBUTES AND INSTITUTIONAL LEARNING OUTCOMES (ILOs)**



| Graduate Attribute (GA) | Descriptors (Institutional Learning Outcome)[1] |
|---|---|
| 1. God-centered | a) Creates an environment where the experience of God is lived and shared |
| | b) Practices honesty, fairness, truth, and integrity in all aspects of life (personal and professional lives) |
| | c) Observes and maintains ethical standards in dealing with the different stakeholders |
| | d) Integrates Christian perspectives and values in all undertakings |
| | e) Manifests humility and respect in relating with other people |
| 2. Patriotic | a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives |
| | b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts |
| | c) Engages actively in political, social, economic, and cultural transformation for nation building |
| | d) Brings pride and honor to the community and the country |
| | e) Patronizes locally produced products and promotes them globally |
| 3. Socially Responsive | a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation |
| | b) Engages in ecological advocacies as responsible stewards of God's creation |
| | c) Continuously works in solidarity with people and institutions to effect liberating action |
| | d) Brings forth awareness and deeper understanding of social realities |
| | e) Manifests spirit of volunteerism through social actions that improve communities |
| 4. Globally Competitive | a) Takes progressive responsibility for own learning and development |
| | b) Promotes critical and creative thinking, self-knowledge and self-mastery |
| | c) Is committed to rise above mediocrity |
| | d) Shows dedication, commitment and love for work to attain exceptional results |
| | e) Exhibits innovativeness and creativity in various contexts |

[1] *Approved in July 2017 during a multi-sectoral committee specifically convened for the purpose of coming up with descriptions for the graduate attributes.*